Predicting the effects of behavioral interventions on humans with LLMs

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We test the potential of predicting human behavior in survey experiments with behavioral clones—large language model (LLM) generated artificial agents mimicking human participants. To evaluate this potential, we predict the outcomes of a large-scale field experiment (N = ~22,000) to be conducted in the United States. The experiment will test 20 text-based interventions designed to increase trust in climate scientists, assessing a broad range of secondary, outcomes including self-reported attitudes (e.g., support for funding of climate science) and behavioral measures (e.g., donations to a scientific organization). Before the human study is fielded, we will simulate it with behavioral clones using demographically representative U.S. participant profiles. We will lock the simulated dataset and analysis code. Once human data are available, we will compare clone-based predictions to human results using a set of preregistered metrics spanning (i) inferential agreement and estimate convergence on treatment effects, (ii) outcome distributions, and (iii) subgroup heterogeneity. Our study will provide a large-scale, credible benchmark test of LLMs' potential—and limitations—as tools for simulating human subjects in social science research. All analyses in this preregistration are demonstrated on simulated placeholder data with identical structure to the real datasets; no human outcome data have been observed at the time of registration.

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```r
# packages
library(tidyverse)
library(sandwich)
library(lmtest)
library(marginaleffects)
library(broom)
library(MetBrewer)
library(patchwork)
library(gt)
library(here)

# custom functions
source(here("R/functions/statistics.R"))
source(here("R/functions/plots.R"))
source(here("R/functions/tables.R"))
source(here("R/functions/reporting.R"))

# simulated (random) data
human_data <- readRDS(here("data/simulation_preregistration/human_data_preregistration.rds"))
llm_data   <- readRDS(here("data/simulation_preregistration/llm_data_preregistration.rds"))

# actual data later
# human_data  <- readRDS("data/cleaned_llm.rds")
```

```
# llm_data <- readRDS("data/cleaned_llm.rds")
```

> **!** Important
>
> **Placeholder data.** Both datasets used throughout are simulated random noise with the correct structure. The analysis code will be applied to real human and LLM clone data without modification once both are available.

Research Questions

The main outcome of the megastudy is trust in climate scientists. However, there are several secondary and tertiary outcomes. For each of these outcomes we have the following research questions:

RQ1 — ATE recovery. How well do LLM clones predict the average treatment effects (ATEs) across 20 interventions?

RQ2 — Distribution shape recovery. How well do LLM clones mirror human response distributions (within experimental conditions)?

RQ3 — Subgroup heterogeneity recovery. How well do LLM clones recover differences in intervention effects across different demographic groups?

RQ4 — Individual-level demographic calibration. Are LLM clone responses calibrated to the demographic profile of each simulated participant? We examine (i) whether clones reproduce the correct baseline response level for each demographic group, (ii) whether demographic variables predict clone responses with the same magnitude as in humans (stereotyping test), and (iii) whether within-subgroup response distributions match between humans and clones.

Secondary — Stacked-model discrepancy test. As a complement to the TOST-based equivalence tests in RQ1, we pool human and clone data and estimate a condition × source interaction, providing a direct inferential test of whether each clone ATE significantly departs from its human counterpart.

Measures

```
outcomes_primary <- "trust_multidimensional"

outcomes_secondary <- c(
  "trust_post", "distrust_post",
```

```r
  #"donation_ams",
  "funding_perceptions", "policy_role_mean", "inst_trust_mean"
)

outcomes_binary <- "newsletter_signup"

outcomes_tertiary <- c(
  "belief_post", "concern_mean", "policy_general",
  "policy_specific_mean", "behavior_mean"
)

outcomes_continuous <- c(outcomes_primary, outcomes_secondary, outcomes_tertiary)

trust_dimensions <- c(
  "trust_competence", "trust_integrity", "trust_benevolence", "trust_openness"
)

trust_items <- c(
  paste0("trust_competence_",  1:3),
  paste0("trust_integrity_",   1:3),
  paste0("trust_benevolence_", 1:3),
  paste0("trust_openness_",    1:3)
)

moderators_cat <- c("gender", "age_band", "race", "education", "income", "party")

outcome_label_map <- c(
  trust_multidimensional = "Multidimensional trust (primary)",
  trust_post             = "Overall trust",
  distrust_post          = "Distrust",
  donation_ams           = "Donation to AMS ($)",
  newsletter_signup      = "Newsletter sign-up",
  funding_perceptions    = "Funding perceptions",
  policy_role_mean       = "Policy role",
  inst_trust_mean        = "Institutional trust",
  belief_post            = "Climate belief",
  concern_mean           = "Climate concern",
  policy_general         = "General climate policy",
  policy_specific_mean   = "Specific climate policy",
  behavior_mean          = "Pro-climate behavior"
)
```

Comparison Functions

Two helper functions operationalize the comparison metrics. They are applied symmetrically to human and LLM clone model outputs throughout the document.

Comparison metrics

`compare_estimates()` is used throughout the document — for ATEs (RQ1), moderator interaction estimates (RQ3), and secondary analyses. It takes two tidy model result tibbles and a `join_by` argument specifying the key columns, and computes six metrics:

- **Spearman** — rank correlation of point estimates; primary metric for RQ1
- **Pearson r** — linear correlation of point estimates
- **RMSE** — root mean squared error; captures magnitude accuracy
- **Directional agreement** — % of estimates with the same sign
- **Inferential agreement** — % of estimates with the same BH-adjusted conclusion (significantly positive, significantly negative, or non-significant at = .05)
- **TOST (%)** — % of individual estimate pairs that pass an equivalence test with preregistered bound $\Delta_k = 0.5 \times |\text{estimate_human_k}|$; equivalence is declared (two one-sided tests, = .05) when the clone estimate falls within ±50% of the human estimate. Note: when human estimates are near zero, Δ_k shrinks accordingly, making equivalence harder to achieve.

```r
compare_estimates <- function(h_result, l_result,
                              join_by = "condition",
                              tost_delta_pct = 0.5) {

  # Standardize columns; SE derived from std.error or from 95% CI width if absent
  prep <- function(df, suffix) {
    df |>
      select(all_of(join_by), estimate) |>
      mutate(
        se    = if ("std.error" %in% names(df)) df$std.error
                else if (all(c("conf.low", "conf.high") %in% names(df)))
                  (df$conf.high - df$conf.low) / (2 * 1.96)
                else NA_real_,
        p_adj = if ("p.value_adjusted" %in% names(df)) df$p.value_adjusted
                else NA_real_
      ) |>
      rename_with(~ paste0(.x, "_", suffix), c(estimate, se, p_adj))
  }
```

```r
  joined <- inner_join(prep(h_result, "h"), prep(l_result, "l"), by = join_by)

  # Three-level inferential category based on BH-adjusted p and direction
  infer_cat <- function(est, p_adj) {
    case_when(
      p_adj < 0.05 & est > 0 ~ "sig_positive",
      p_adj < 0.05 & est < 0 ~ "sig_negative",
      TRUE                   ~ "not_significant"
    )
  }

  # TOST: equivalence declared when |ATE_h - ATE_l| < delta = tost_delta_pct × |ATE_h|
  # p_tost is the maximum of the two one-sided p-values; equivalent if p_tost < 0.05
  joined |>
    mutate(
      delta      = tost_delta_pct * abs(estimate_h),
      diff       = estimate_h - estimate_l,
      se_diff    = sqrt(se_h^2 + se_l^2),
      p_lower    = pnorm((diff + delta) / se_diff, lower.tail = FALSE),
      p_upper    = pnorm((diff - delta) / se_diff),
      p_tost     = pmax(p_lower, p_upper),
      equivalent = p_tost < 0.05,
      same_infer = infer_cat(estimate_h, p_adj_h) == infer_cat(estimate_l, p_adj_l)
    ) |>
    summarise(
      spearman_rho    = cor(estimate_h, estimate_l, method = "spearman",
                            use = "pairwise.complete.obs"),
      pearson_r       = cor(estimate_h, estimate_l, use = "pairwise.complete.obs"),
      rmse            = sqrt(mean((estimate_h - estimate_l)^2, na.rm = TRUE)),
      directional_pct = mean(sign(estimate_h) == sign(estimate_l), na.rm = TRUE) * 100,
      inferential_pct = if (!anyNA(p_adj_h) && !anyNA(p_adj_l))
        mean(same_infer, na.rm = TRUE) * 100 else NA_real_,
      tost_pct        = mean(equivalent, na.rm = TRUE) * 100
    )
}
```

Distribution comparison metrics

`compare_distributions()` compares response distributions for a single condition and outcome using three metrics:

- **OVL (overlapping coefficient)** — proportion of area where the two kernel density estimates overlap; 0 = no overlap, 1 = identical distributions; primary metric for RQ2
- **KS D-statistic** — maximum absolute difference between the two empirical CDFs; 0 = identical
- **Variance ratio** — LLM clone variance divided by human variance; 1 = equal spread

```
compute_ovl <- function(x, y, n_grid = 512) {
  lo <- min(c(x, y)); hi <- max(c(x, y))
  if (lo == hi) return(NA_real_)
  # Evaluate both KDEs on the same grid, then integrate the overlap area
  d_h <- density(x, from = lo, to = hi, n = n_grid)
  d_l <- density(y, from = lo, to = hi, n = n_grid)
  sum(pmin(d_h$y, d_l$y)) * (hi - lo) / n_grid
}

compare_distributions <- function(human_data, llm_data, outcome, condition_val) {
  x <- human_data |> filter(condition == condition_val) |> pull(all_of(outcome)) |> na.omit(
  y <- llm_data   |> filter(condition == condition_val) |> pull(all_of(outcome)) |> na.omit(
  tibble(
    condition      = as.character(condition_val),
    ovl            = compute_ovl(x, y),
    ks_d           = suppressWarnings(ks.test(x, y)$statistic),
    variance_ratio = var(y) / var(x)  # > 1: clones more variable; < 1: less variable
  )
}
```

Calibration regression

`run_calibration()` regresses human ATEs on LLM clone ATEs: `ATE_human =  +  × ATE_clone`. The intercept captures systematic bias (positive means clones underestimate the baseline, negative means overestimation). The slope captures proportionality: = 1 is perfect calibration, > 1 means clone effects are too small relative to human effects, < 1 means they are too large. We report a formal F-test of = 1 alongside , , and R^2.

```
run_calibration <- function(h_result, l_result) {
  joined <- inner_join(
    h_result |> select(condition, estimate_h = estimate),
    l_result |> select(condition, estimate_l = estimate),
    by = "condition"
  )

  # ATE_h =  +  × ATE_l; perfect calibration:  = 0 and  = 1
```

```r
  fit   <- lm(estimate_h ~ estimate_l, data = joined)
  coefs <- broom::tidy(fit, conf.int = TRUE)

  tibble(
    alpha     = coefs$estimate[1],
    alpha_lo  = coefs$conf.low[1],
    alpha_hi  = coefs$conf.high[1],
    beta      = coefs$estimate[2],
    beta_lo   = coefs$conf.low[2],
    beta_hi   = coefs$conf.high[2],
    r_squared = broom::glance(fit)$r.squared,
    # F-test H0:   = 1 (clone ATEs are proportionally calibrated to human ATEs)
    p_beta_1  = car::linearHypothesis(fit, "estimate_l = 1")$`Pr(>F)`[2]
  )
}
```

RQ1: Average Treatment Effect Recovery

For each preregistered outcome we run `run_main_treatment_model()` (OLS, HC2 robust SEs, BH-adjusted p-values) on both datasets separately and compare the 20 intervention-vs-control ATEs using `compare_estimates()`. For `newsletter_signup` (binary), we use marginal effects from `run_main_treatment_model_binary()`.

Primary outcome

```r
h_primary <- run_main_treatment_model(human_data, outcome = "trust_multidimensional")
l_primary <- run_main_treatment_model(llm_data,   outcome = "trust_multidimensional")

compare_estimates(h_primary, l_primary)
```

```
# A tibble: 1 x 6
  spearman_rho pearson_r  rmse directional_pct inferential_pct tost_pct
         <dbl>     <dbl> <dbl>           <dbl>           <dbl>    <dbl>
1       0.0226    0.0324 0.408              50             100        0
```

```r
inner_join(
  h_primary |> select(condition, estimate_h = estimate),
  l_primary |> select(condition, estimate_l = estimate),
  by = "condition"
```

```
) |>
  ggplot(aes(x = estimate_h, y = estimate_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.5) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_point(size = 2.5, alpha = 0.8, color = "grey20") +
  geom_text(aes(label = str_remove(condition, "intervention_")),
            size = 2.5, vjust = -0.8, color = "grey40") +
  labs(x = "Human ATE (OLS estimate)", y = "LLM clone ATE (OLS estimate)") +
  plot_theme
```

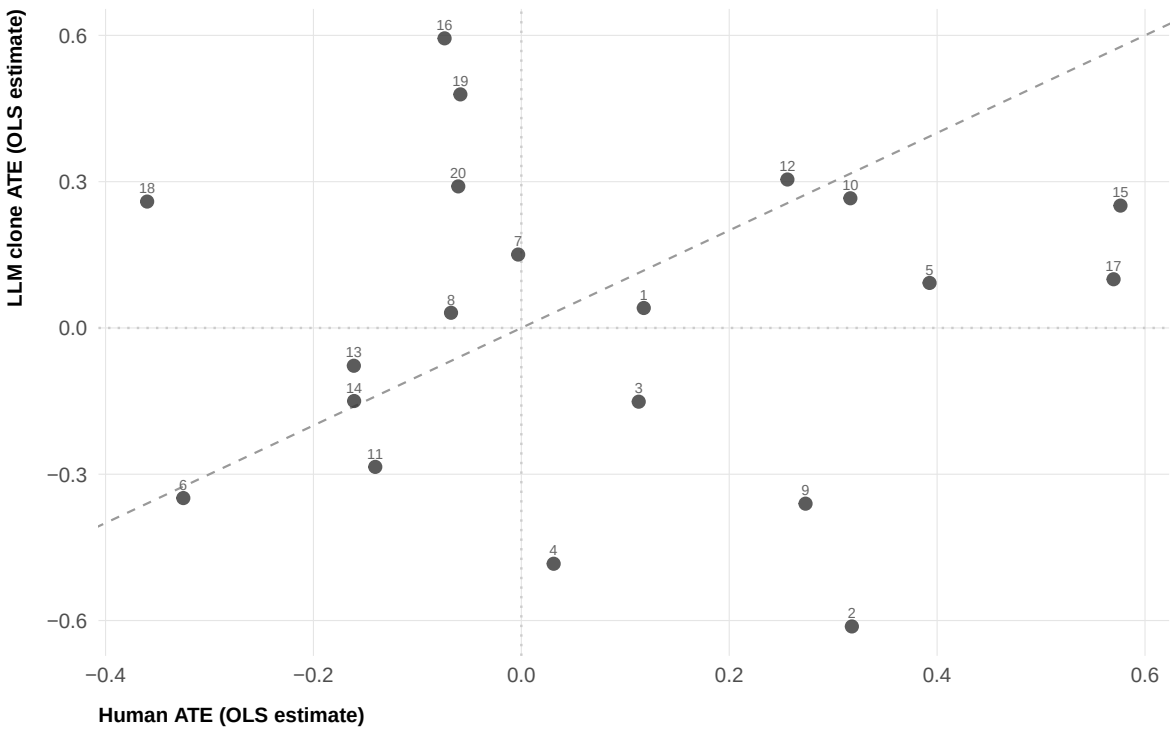

Figure 1: Human vs. LLM clone ATEs for the primary outcome (multidimensional trust). Each point is one of the 20 interventions. The dashed line indicates perfect agreement; dotted lines mark zero.

```
run_calibration(h_primary, l_primary)
```

```
# A tibble: 1 x 8
```

```
   alpha alpha_lo alpha_hi   beta beta_lo beta_hi r_squared p_beta_1
   <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>     <dbl>    <dbl>
1 0.0771  -0.0533    0.208 0.0270  -0.386   0.440   0.00105 0.000103
```

All outcomes

```r
# Run and store models to avoid re-fitting for calibration
ate_model_results <- map(outcomes_continuous, function(out) {
  list(
    h = run_main_treatment_model(human_data, outcome = out),
    l = run_main_treatment_model(llm_data,   outcome = out)
  )
}) |> set_names(outcomes_continuous)

ate_metrics <- map_dfr(outcomes_continuous, function(out) {
  compare_estimates(ate_model_results[[out]]$h, ate_model_results[[out]]$l) |>
    mutate(outcome = out)
})

calib_metrics <- map_dfr(outcomes_continuous, function(out) {
  run_calibration(ate_model_results[[out]]$h, ate_model_results[[out]]$l) |>
    mutate(outcome = out)
})

# Binary: newsletter signup (marginal effects on probability scale)
h_bin <- run_main_treatment_model_binary(human_data, "newsletter_signup")
l_bin <- run_main_treatment_model_binary(llm_data,   "newsletter_signup")

ate_metrics <- bind_rows(
  ate_metrics,
  compare_estimates(h_bin$marginal_effects, l_bin$marginal_effects) |>
    mutate(outcome = "newsletter_signup")
) |>
  mutate(
    tier = case_when(
      outcome == outcomes_primary                          ~ "Primary",
      outcome %in% c(outcomes_secondary, outcomes_binary) ~ "Secondary",
      TRUE                                                 ~ "Tertiary"
    ),
    outcome_label = outcome_label_map[outcome]
  )
```

```r
calib_metrics <- bind_rows(
  calib_metrics,
  run_calibration(h_bin$marginal_effects, l_bin$marginal_effects) |>
    mutate(outcome = "newsletter_signup")
) |>
  mutate(
    tier = case_when(
      outcome == outcomes_primary                          ~ "Primary",
      outcome %in% c(outcomes_secondary, outcomes_binary) ~ "Secondary",
      TRUE                                                 ~ "Tertiary"
    ),
    outcome_label = outcome_label_map[outcome]
  )
```

```r
ate_metrics |>
  arrange(match(tier, c("Primary", "Secondary", "Tertiary")), outcome_label) |>
  select(Tier = tier, Outcome = outcome_label,
         `Spearman  ` = spearman_rho, `Pearson r` = pearson_r, RMSE = rmse,
         `Directional (%)` = directional_pct, `Inferential (%)` = inferential_pct,
         `TOST (%)` = tost_pct) |>
  gt(groupname_col = "Tier") |>
  fmt_number(columns = c(`Spearman  `, `Pearson r`, RMSE), decimals = 3) |>
  fmt_number(columns = c(`Directional (%)`, `Inferential (%)`, `TOST (%)`), decimals = 1) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_row_groups()
  )
```

Table 1: RQ1 comparison metrics by outcome. Spearman and Pearson r: correlations of the 20 ATEs. RMSE: root mean squared error. Directional and inferential agreement are percentages out of 20 interventions. TOST: % of ATE pairs achieving equivalence (Δ = 50% of human ATE).

Outcome	Spearman	Pearson r	RMSE	Directional (%)	Inferential (%)
Primary					
Multidimensional trust (primary)	0.023	0.032	0.408	50.0	100.0
Secondary					
Distrust	-0.077	-0.161	1.450	70.0	90.0
Funding perceptions	0.017	-0.009	1.618	50.0	95.0
Institutional trust	-0.205	-0.140	1.224	30.0	100.0
Newsletter sign-up	0.266	0.223	0.016	65.0	100.0
Overall trust	0.096	0.135	1.257	60.0	100.0
Policy role	-0.122	-0.127	0.700	30.0	100.0
Tertiary					
Climate belief	-0.117	-0.035	1.395	45.0	100.0
Climate concern	0.021	0.099	1.140	40.0	100.0
General climate policy	-0.005	0.026	1.345	60.0	100.0
Pro-climate behavior	-0.409	-0.477	1.234	40.0	50.0
Specific climate policy	0.301	0.279	0.377	65.0	100.0

```
calib_metrics |>
  arrange(match(tier, c("Primary", "Secondary", "Tertiary")), outcome_label) |>
  mutate(
    beta_ci  = paste0("[", round(beta_lo,  2), ", ", round(beta_hi,  2), "]"),
    alpha_ci = paste0("[", round(alpha_lo, 2), ", ", round(alpha_hi, 2), "]")
  ) |>
  select(Tier = tier, Outcome = outcome_label,
         alpha, alpha_ci, beta, beta_ci, r_squared, p_beta_1) |>
  gt(groupname_col = "Tier") |>
  cols_label(alpha = " ", alpha_ci = "95% CI ( )",
             beta = " ", beta_ci = "95% CI ( )",
             r_squared = "R²", p_beta_1 = "p ( =1)") |>
  fmt_number(columns = c(alpha, beta, r_squared), decimals = 3) |>
  fmt_number(columns = p_beta_1, decimals = 3) |>
  tab_style(
    style     = cell_text(weight = "bold"),
```

```
    locations = cells_row_groups()
  )
```

Table 2: Calibration regression by outcome. : intercept (systematic bias, with 95% CI). : slope (proportionality; 1 = perfect, with 95% CI). R^2: variance explained. p(=1): F-test of perfect calibration.

Outcome		95% CI ( )		95% CI ( )	R^2	p (=1)
Primary						
Multidimensional trust (primary)	0.077	[-0.05, 0.21]	0.027	[-0.39, 0.44]	0.001	0.000
Secondary						
Distrust	1.043	[0.31, 1.78]	-0.194	[-0.78, 0.39]	0.026	0.000
Funding perceptions	0.455	[-0.18, 1.09]	-0.014	[-0.79, 0.76]	0.000	0.013
Institutional trust	0.184	[-0.27, 0.64]	-0.154	[-0.7, 0.39]	0.020	0.000
Newsletter sign-up	0.009	[0, 0.02]	0.170	[-0.2, 0.54]	0.050	0.000
Overall trust	0.272	[-0.23, 0.77]	0.140	[-0.37, 0.65]	0.018	0.002
Policy role	-0.074	[-0.27, 0.12]	-0.093	[-0.45, 0.27]	0.016	0.000
Tertiary						
Climate belief	0.557	[-0.02, 1.14]	-0.048	[-0.72, 0.63]	0.001	0.004
Climate concern	-0.407	[-0.79, -0.03]	0.101	[-0.4, 0.6]	0.010	0.001
General climate policy	-0.225	[-0.71, 0.26]	0.020	[-0.37, 0.41]	0.001	0.000
Pro-climate behavior	0.897	[0.7, 1.09]	-0.672	[-1.29, -0.06]	0.228	0.000
Specific climate policy	0.039	[-0.1, 0.17]	0.238	[-0.17, 0.64]	0.078	0.001

```
map_dfr(outcomes_continuous, function(out) {
  h <- run_main_treatment_model(human_data, outcome = out)
  l <- run_main_treatment_model(llm_data,   outcome = out)
  inner_join(
    h |> select(condition, estimate_h = estimate),
    l |> select(condition, estimate_l = estimate),
    by = "condition"
  ) |> mutate(outcome = outcome_label_map[out])
}) |>
  ggplot(aes(x = estimate_h, y = estimate_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.4) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
```

```r
geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
geom_point(size = 1.5, alpha = 0.7, color = "grey20") +
facet_wrap(~ outcome, scales = "free", ncol = 3) +
labs(x = "Human ATE", y = "LLM clone ATE") +
plot_theme
```

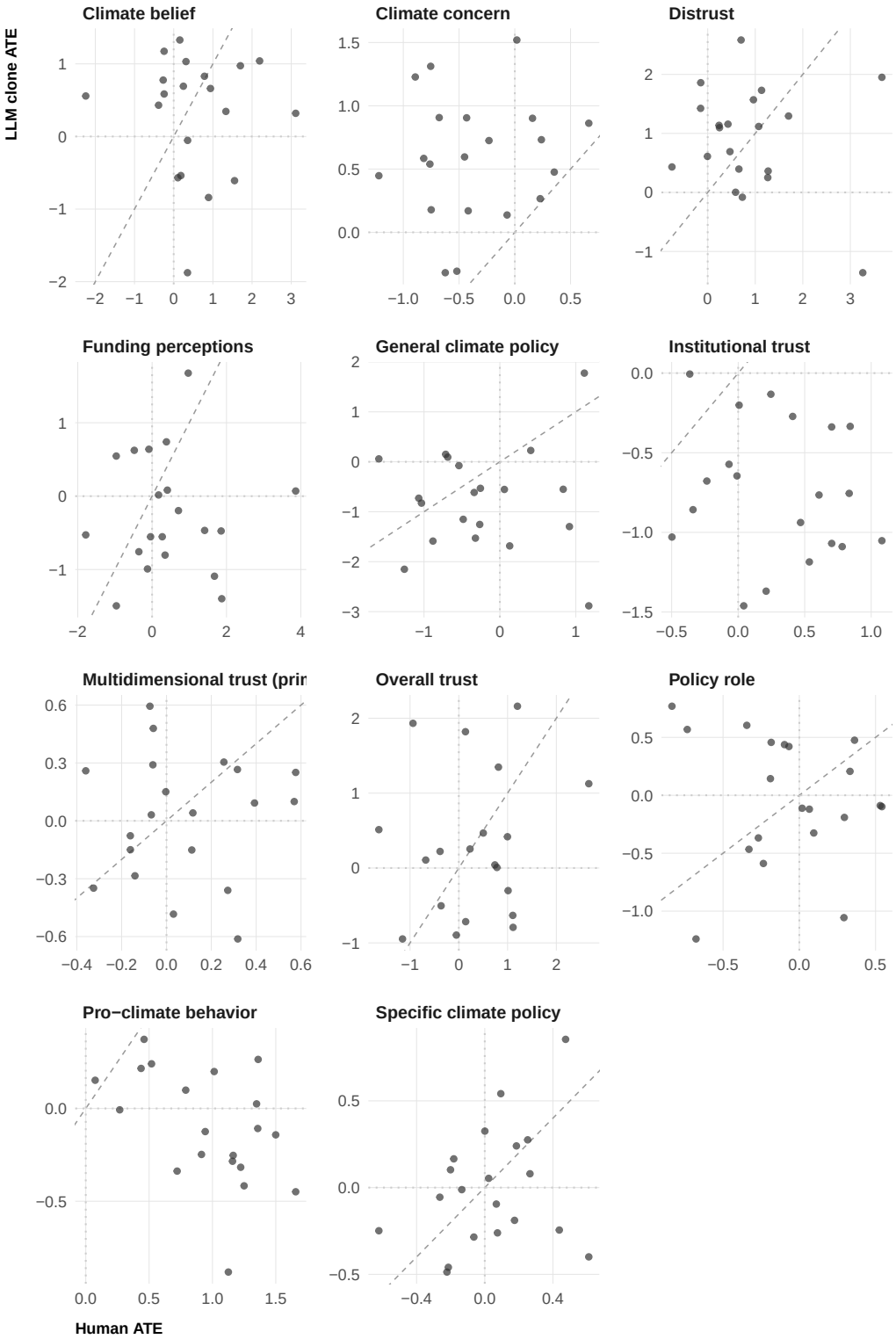

Figure 2: Human vs. LLM clone ATEs for all continuous outcomes. Each point is one of the 20 interventions; axes are free-scaled across panels.

Secondary: Stacked-model discrepancy test

As a complement to the TOST equivalence tests above, we pool both datasets and estimate a `condition × source` interaction. The interaction term for each intervention directly estimates the human–clone ATE discrepancy and provides a conventional (directional) inferential test alongside the equivalence tests.

```
# Stack and test condition × source interaction for the primary outcome.
# Positive estimate_h > estimate_l → clone underestimates; negative → overestimates.
stacked_primary <- run_stacked_model(human_data, llm_data,
                                     outcome = "trust_multidimensional")
```

```
stacked_primary |>
  select(Condition = condition, Estimate = estimate,
         `95% CI low` = conf.low, `95% CI high` = conf.high,
         `p (adj)` = p.value_adjusted, ` ` = significant_adjusted) |>
  gt() |>
  fmt_number(columns = c(Estimate, `95% CI low`, `95% CI high`), decimals = 3) |>
  fmt_number(columns = `p (adj)`, decimals = 3)
```

Table 3: Stacked-model discrepancy estimates for the primary outcome. Each row is one intervention; estimate = ATE_human − ATE_clone from the condition × source interaction.

Condition	Estimate	95% CI low	95% CI high	p (adj)	
intervention_1	-0.077	-0.973	0.819	0.981	NA
intervention_10	-0.051	-0.947	0.846	0.981	NA
intervention_11	-0.144	-1.052	0.764	0.981	NA
intervention_12	0.048	-0.870	0.966	0.981	NA
intervention_13	0.084	-0.816	0.983	0.981	NA
intervention_14	0.011	-0.892	0.914	0.981	NA
intervention_15	-0.325	-1.224	0.573	0.981	NA
intervention_16	0.668	-0.245	1.580	0.855	NA
intervention_17	-0.470	-1.357	0.417	0.855	NA
intervention_18	0.619	-0.286	1.524	0.855	NA
intervention_19	0.538	-0.348	1.423	0.855	NA
intervention_2	-0.930	-1.829	-0.031	0.852	NA
intervention_20	0.351	-0.537	1.238	0.981	NA
intervention_3	-0.264	-1.168	0.639	0.981	NA
intervention_4	-0.515	-1.410	0.380	0.855	NA
intervention_5	-0.300	-1.196	0.596	0.981	NA
intervention_6	-0.024	-0.924	0.876	0.981	NA
intervention_7	0.154	-0.756	1.063	0.981	NA
intervention_8	0.099	-0.801	0.998	0.981	NA
intervention_9	-0.634	-1.540	0.273	0.855	NA

RQ2: Response Distribution Shape Recovery

For the primary outcome (`trust_multidimensional`), we compare response distributions of human and LLM clone participants within each of the 21 experimental conditions (control + 20 interventions).

```
conditions_all <- levels(human_data$condition)

dist_metrics <- map_dfr(conditions_all, function(cond) {
  compare_distributions(human_data, llm_data, "trust_multidimensional", cond)
})
```

```r
source_colors <- c("Human" = "grey30", "LLM clone" = met.brewer("Juarez", n = 10)[1])

bind_rows(
  human_data |> select(condition, value = trust_multidimensional) |>
    mutate(source = "Human"),
  llm_data   |> select(condition, value = trust_multidimensional) |>
    mutate(source = "LLM clone")
) |>
  filter(!is.na(value)) |>
  ggplot(aes(x = value, fill = source, color = source)) +
  geom_density(alpha = 0.35, linewidth = 0.4) +
  scale_fill_manual(values  = source_colors, name = NULL) +
  scale_color_manual(values = source_colors, name = NULL) +
  facet_wrap(~ condition, ncol = 4) +
  labs(x = "Multidimensional trust (0-100)", y = "Density") +
  plot_theme +
  theme(legend.position = "top")
```

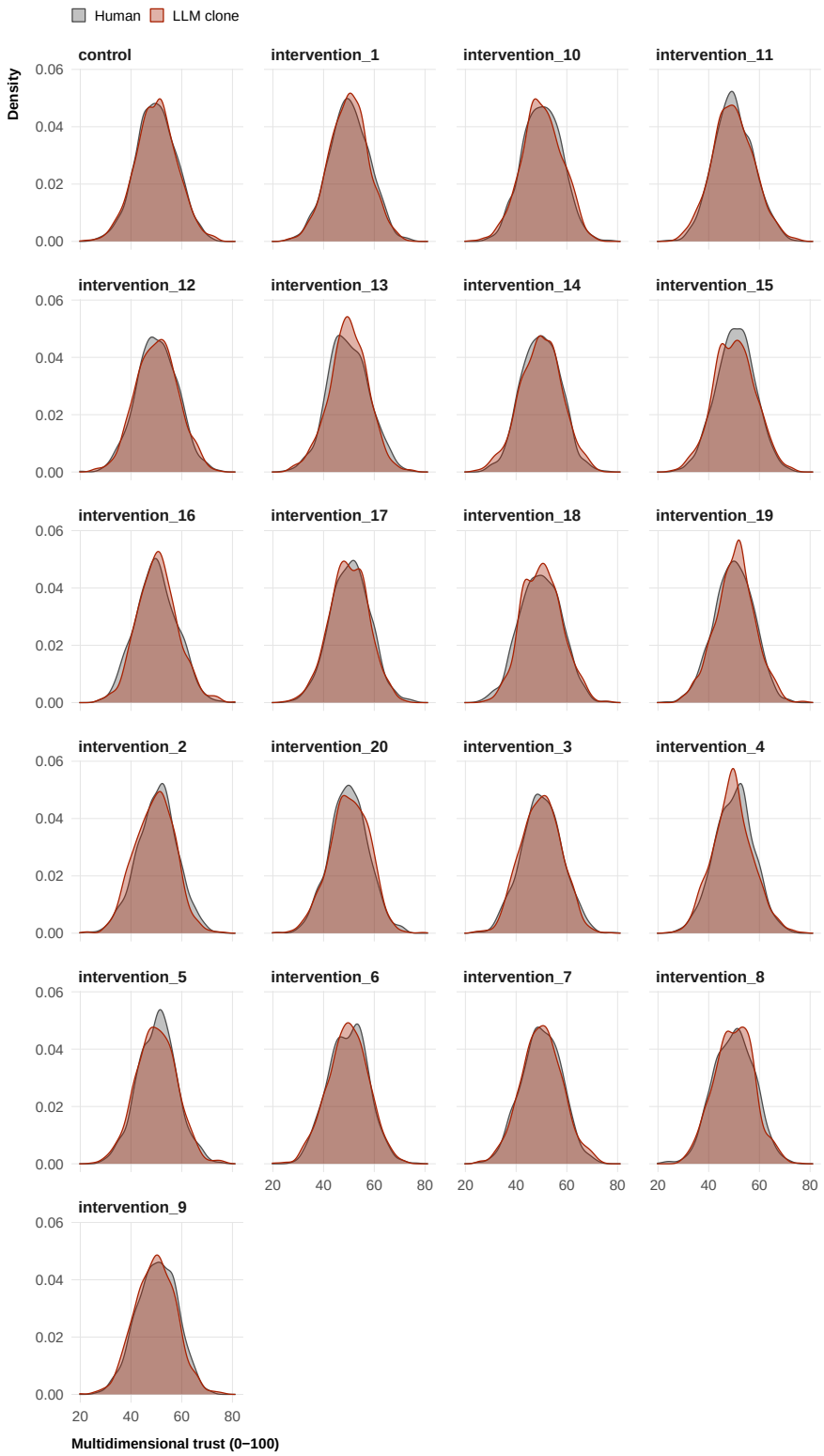

Figure 3: Response distributions for multidimensional trust by experimental condition. Human participants shown in dark grey, LLM clones in orange.

```r
dist_metrics |>
  mutate(condition = factor(condition, levels = conditions_all)) |>
  arrange(condition) |>
  mutate(condition = as.character(condition)) |>
  gt() |>
  cols_label(condition = "Condition", ovl = "OVL",
             ks_d = "KS D", variance_ratio = "Variance ratio") |>
  fmt_number(columns = c(ovl, ks_d, variance_ratio), decimals = 3)
```

Table 4: RQ2 distribution comparison metrics by condition. OVL: overlapping coefficient (1 = identical). KS D: Kolmogorov–Smirnov D-statistic (0 = identical CDFs). Variance ratio: LLM clone / human variance.

Condition	OVL	KS D	Variance ratio
control	0.974	0.016	1.051
intervention_1	0.962	0.043	0.931
intervention_10	0.953	0.028	1.069
intervention_11	0.967	0.038	1.115
intervention_12	0.961	0.026	1.030
intervention_13	0.930	0.053	0.923
intervention_14	0.961	0.025	1.150
intervention_15	0.938	0.070	1.143
intervention_16	0.950	0.044	0.967
intervention_17	0.963	0.039	0.969
intervention_18	0.950	0.048	0.919
intervention_19	0.951	0.047	1.005
intervention_2	0.942	0.061	0.967
intervention_20	0.949	0.049	1.061
intervention_3	0.966	0.039	0.956
intervention_4	0.931	0.089	1.055
intervention_5	0.953	0.047	1.090
intervention_6	0.959	0.027	1.067
intervention_7	0.965	0.021	1.058
intervention_8	0.949	0.040	0.933
intervention_9	0.952	0.052	1.047

Mean OVL across all conditions: 0.954. Mean KS D: 0.043. Mean variance ratio: 1.024.

RQ3: Subgroup Heterogeneity Recovery

For each of six categorical moderators, we run `run_moderator_model()` on both datasets for the primary outcome and compare the condition × moderator interaction estimates using `compare_estimates()` (with `join_by = c("condition", "moderator_level")`).

```
mod_results_h <- map(set_names(moderators_cat), function(mod) {
  run_moderator_model(human_data, outcome = "trust_multidimensional", moderator = mod)
})

mod_results_l <- map(set_names(moderators_cat), function(mod) {
  run_moderator_model(llm_data, outcome = "trust_multidimensional", moderator = mod)
})
```

```
rq3_metrics <- map_dfr(moderators_cat, function(mod) {
  compare_estimates(
    mod_results_h[[mod]]$interaction_effects,
    mod_results_l[[mod]]$interaction_effects,
    join_by = c("condition", "moderator_level")
  ) |> mutate(
    moderator   = mod,
    n_estimates = nrow(mod_results_h[[mod]]$interaction_effects)
  )
})
```

```
rq3_metrics |>
  select(Moderator = moderator, `Spearman  ` = spearman_rho, `Pearson r` = pearson_r,
         RMSE = rmse, `Directional (%)` = directional_pct,
         `Inferential (%)` = inferential_pct, `TOST (%)` = tost_pct,
         `N estimates` = n_estimates) |>
  gt() |>
  fmt_number(columns = c(`Spearman  `, `Pearson r`, RMSE), decimals = 3) |>
  fmt_number(columns = c(`Directional (%)`, `Inferential (%)`, `TOST (%)`), decimals = 1)
```

Table 5: RQ3 interaction estimate comparison metrics by moderator. Spearman : rank correlation across all condition × moderator level interaction estimates. Directional agreement: % with the same sign. N: number of interaction estimates compared.

Moderator	Spearman	Pearson r	RMSE	Directional (%)	Inferential (%)	TOST (%)	N estim
gender	0.280	0.290	1.042	62.5	100.0	0.0	
age_band	0.085	0.051	1.449	57.0	100.0	0.0	
race	0.091	0.132	1.737	50.0	100.0	0.0	
education	-0.100	-0.077	1.470	40.0	100.0	0.0	
income	-0.161	-0.192	1.488	51.2	100.0	0.0	
party	-0.067	-0.040	1.175	53.3	100.0	0.0	

```
map_dfr(moderators_cat, function(mod) {
  inner_join(
    mod_results_h[[mod]]$interaction_effects |>
      select(condition, moderator_level, estimate_h = estimate),
    mod_results_l[[mod]]$interaction_effects |>
      select(condition, moderator_level, estimate_l = estimate),
    by = c("condition", "moderator_level")
  ) |> mutate(moderator = mod)
}) |>
  ggplot(aes(x = estimate_h, y = estimate_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.4) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_point(size = 1.5, alpha = 0.65, color = "grey20") +
  facet_wrap(~ moderator, scales = "free", ncol = 2) +
  labs(x = "Human interaction estimate", y = "LLM clone interaction estimate") +
  plot_theme
```

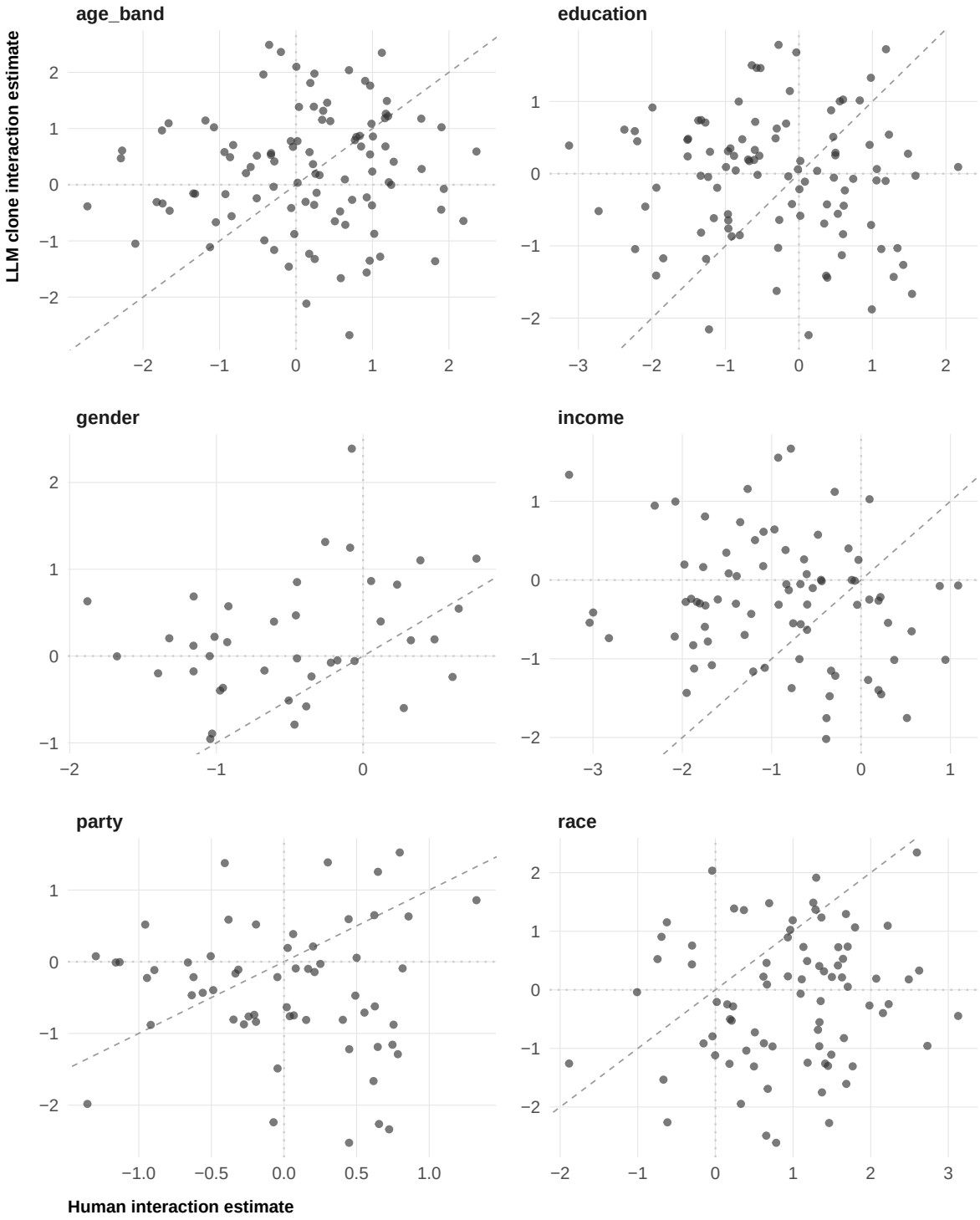

Figure 4: Human vs. LLM clone condition × moderator interaction estimates for the primary outcome, by moderator. Each point is one condition × moderator level combination.

RQ4: Individual-level Demographic Calibration

RQ1–3 operate at the level of aggregated estimates (ATEs, interaction effects, distributions). RQ4 asks whether clone responses are calibrated at the individual level: do clones assigned a given demographic profile give responses that match what real humans with those demographics give? All analyses below use the primary outcome (`trust_multidimensional`) and the control condition unless stated otherwise.

RQ4a: Demographic baseline calibration

For each of the six moderators, we compute mean outcomes within each demographic group in the control condition separately for humans and clones, then correlate and compare those cell means. High correlation and low RMSE indicate that clones have the right baseline response level for each demographic type.

```
# Cell-mean comparison in the control condition, one moderator at a time
baseline_cal <- compare_demographic_baselines(
  human_data, llm_data,
  outcome    = "trust_multidimensional",
  moderators = moderators_cat
)
```

```
baseline_cal |>
  select(Moderator = moderator, r, RMSE = rmse, `N cells` = n_cells) |>
  gt() |>
  fmt_number(columns = c(r, RMSE), decimals = 3)
```

Table 6: Demographic baseline calibration. For each moderator, cell means in the control condition are compared between humans and clones. r: Pearson correlation of cell means. RMSE: root mean squared error. N cells: number of demographic groups.

Moderator	r	RMSE	N cells
gender	-0.797	0.357	3
age_band	0.468	0.436	6
race	0.302	0.681	5
education	-0.966	0.684	6
income	-0.143	0.530	5
party	-0.731	0.511	4

RQ4b: Demographic predictability — stereotyping test

We regress the primary outcome on all six demographic moderators plus condition fixed effects separately for humans and clones. A substantially higher R^2 in clones indicates over-reliance on demographic stereotypes (mapping demographics to responses too mechanically). We also compare individual demographic coefficients side by side.

```
pred_test <- compare_demographic_predictability(
  human_data, llm_data,
  outcome    = "trust_multidimensional",
  predictors = moderators_cat
)
```

```
pred_test$r_squared |>
  mutate(source = if_else(source == "human", "Human", "LLM clone")) |>
  gt() |>
  cols_label(source = "Source", r_squared = "R²") |>
  fmt_number(columns = r_squared, decimals = 3)
```

Table 7: R^2 from regressing the primary outcome on demographics + condition FE, separately for humans and clones. Clone R^2 » human R^2 signals demographic stereotyping.

Source	R^2
Human	0.002
LLM clone	0.002

```
pred_test$coefficients |>
  ggplot(aes(x = est_h, y = est_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.5) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_point(size = 2, alpha = 0.8, color = "grey20") +
  geom_text(aes(label = term), size = 2.2, vjust = -0.8, color = "grey40") +
  labs(x = "Human coefficient", y = "LLM clone coefficient") +
  plot_theme
```

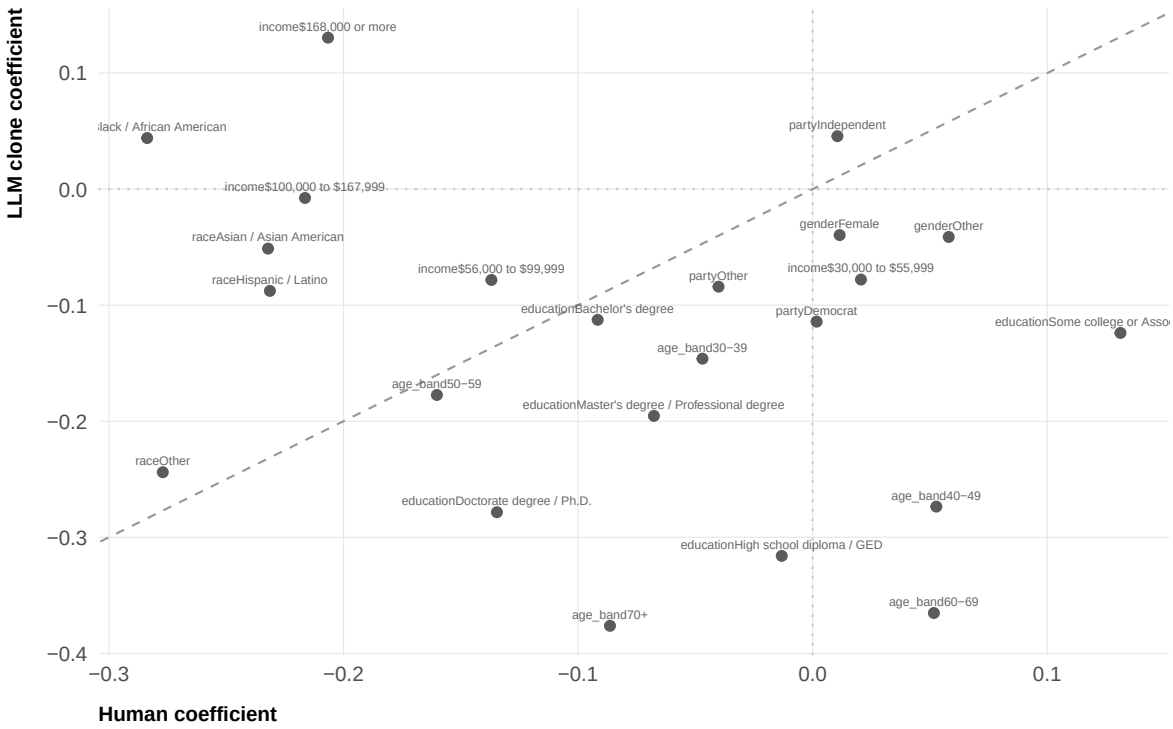

Figure 5: Demographic regression coefficients for the primary outcome in humans (x-axis) vs. LLM clones (y-axis). Each point is one dummy-coded level of a moderator. The dashed line marks perfect agreement.

RQ4c: Within-subgroup distribution shapes

We extend RQ2 by comparing response distributions within demographic subgroups in the control condition. This tests whether distributional calibration holds within groups, not just in the aggregate (a clone dataset could match the overall distribution while having miscalibrated within-group spreads).

```
control_val <- levels(human_data$condition)[1]

# For each moderator × level, compare human and clone distributions in control
subgroup_dist <- map_dfr(moderators_cat, function(mod) {
  groups <- levels(human_data[[mod]])
  map_dfr(groups, function(grp) {
    x <- human_data |>
      filter(condition == control_val, .data[[mod]] == grp) |>
      pull(trust_multidimensional) |> na.omit()
```

```r
    y <- llm_data |>
      filter(condition == control_val, .data[[mod]] == grp) |>
      pull(trust_multidimensional) |> na.omit()
    # Require minimum n for stable KDE
    if (length(x) < 10 | length(y) < 10) return(NULL)
    tibble(moderator = mod, group = grp,
           ovl = compute_ovl(x, y),
           ks_d = suppressWarnings(ks.test(x, y)$statistic),
           variance_ratio = var(y) / var(x))
  })
})
```

```r
subgroup_dist |>
  arrange(moderator, group) |>
  gt(groupname_col = "moderator") |>
  cols_label(group = "Group", ovl = "OVL", ks_d = "KS D",
             variance_ratio = "Variance ratio") |>
  fmt_number(columns = c(ovl, ks_d, variance_ratio), decimals = 3) |>
  tab_style(style = cell_text(weight = "bold"),
            locations = cells_row_groups())
```

Table 8: Within-subgroup distribution comparison in the control condition. OVL: overlapping coefficient (1 = identical). KS D: Kolmogorov–Smirnov statistic. Variance ratio: clone / human.

Group	OVL	KS D	Variance ratio
age_band			
18-29	0.927	0.051	1.253
30-39	0.945	0.063	0.999
40-49	0.963	0.051	0.967
50-59	0.961	0.050	1.130
60-69	0.915	0.068	0.940
70+	0.916	0.071	0.951
education			
Bachelor's degree	0.939	0.056	1.188
Doctorate degree / Ph.D.	0.897	0.117	1.141
High school diploma / GED	0.941	0.067	1.061
Less than high school	0.934	0.046	1.157
Master's degree / Professional degree	0.943	0.060	0.960
Some college or Associate's degree	0.939	0.078	0.885
gender			
Female	0.951	0.050	1.226
Male	0.954	0.049	0.976
Other	0.949	0.043	0.964
income			
$100,000 to $167,999	0.939	0.081	1.183
$168,000 or more	0.933	0.075	0.969
$30,000 to $55,999	0.953	0.048	1.030
$56,000 to $99,999	0.962	0.042	1.041
Less than $30,000	0.939	0.057	1.096
party			
Democrat	0.939	0.058	0.949
Independent	0.963	0.044	1.046
Other	0.944	0.054	1.043
Republican	0.950	0.052	1.153
race			
Asian / Asian American	0.941	0.053	1.209
Black / African American	0.924	0.048	0.811
Hispanic / Latino	0.916	0.107	1.252
Other	0.950	0.044	1.047
White / Caucasian	0.949	0.071	0.968